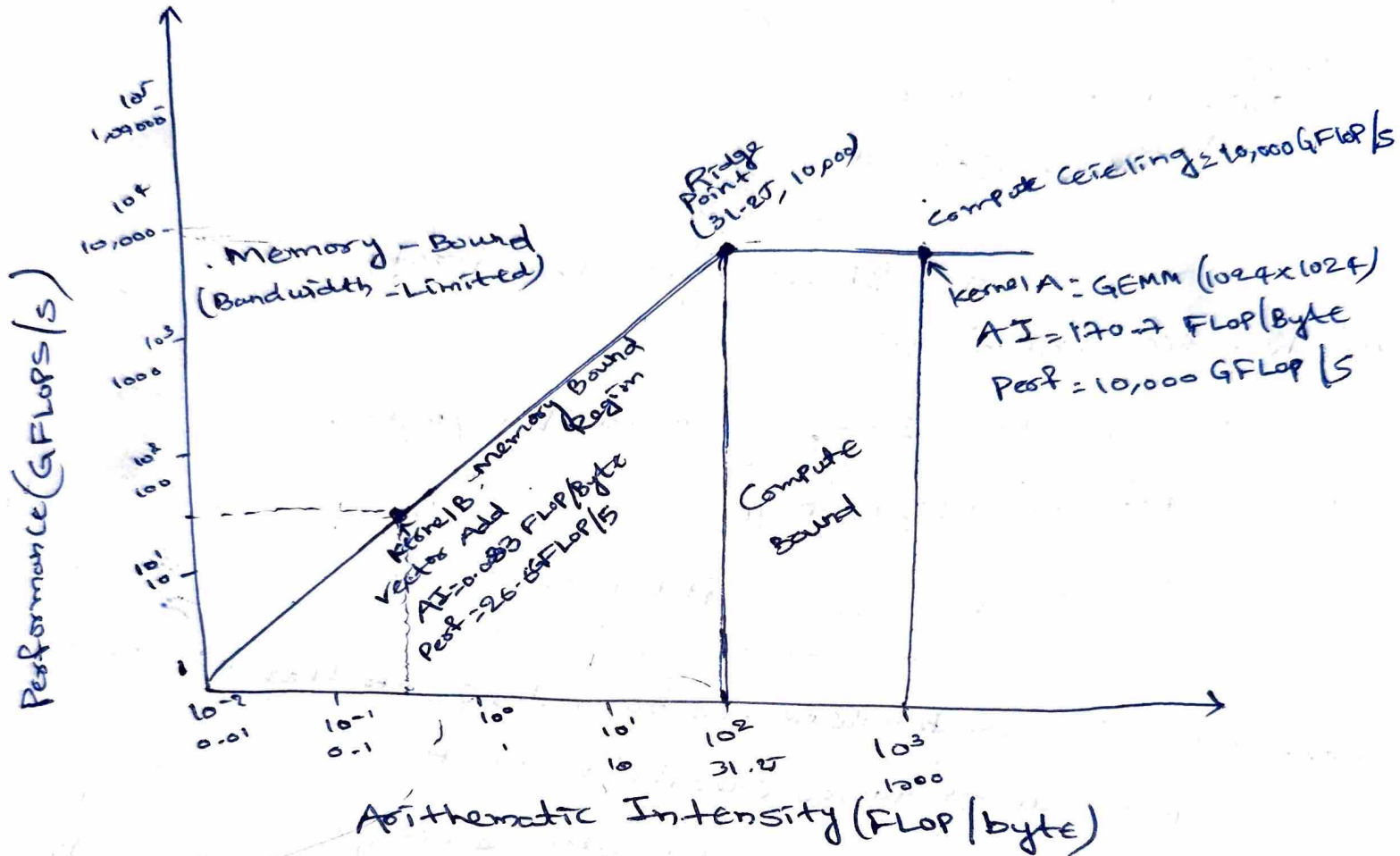


CMAN

Roof line Model 1

a)



Specifications:-

Peak Compute = 10 TFLOPS = 10,000 GFLOPS/s

Peak DRAM Bandwidth = 320 GB/s

Ridge Point = $10,000 / 320 = 31.25$ Flop/byte.

b) Kernel A:-

Arithmetic Intensity: $I = N/6$

$$N = 1024$$

$$I = \frac{1024}{6}$$

$$I = 170.66$$

2) Kernel B:-

$$N = 4,194,304$$

FLOPs:

4,194,304 operations

$$\text{Bytes Transferred} = 3N \times 4$$

$$= 3 \times 4,194,304 \times 4$$

$$= 50,331,648 \text{ bytes transferred.}$$

Arithmetic Intensity:

$$I = N / (N \cdot 3 \times 4) = I = \frac{1}{12} = 0.0833 \text{ FLOPs / Bytes Performance.}$$

$$B_{\text{peak}} = 1320 \text{ GB/s} \times 1/12 \text{ FLOPs/B}$$

$$= 320/12 = \underline{\underline{26.67 \text{ GFLOPs/s}}}$$

d) Kernel A:- It is bottlenecked by math, so the hardware needs more compute units to do calculation faster

Kernel B:- It is bottlenecked by data delivery, so the hardware needs a faster memory connection to keep the processor fed.

→ To improve performance, kernel A requires more processing units because it is limited by math speed, whereas kernel B requires higher memory bandwidth because it is starving for data.